Lesson 1: Move a shape on screen

Pygame CE • S3 onwards • 50-60 min

Learning intention
Understand the game loop and use keyboard input to change an object's position.

Success criteria

- I can create a window and draw a shape.
- I can respond to the arrow keys.
- I can explain how the shape's coordinates change.

Key ideas

- Sketch the window as a grid. **(0, 0)** is the **top-left**; x increases to the right and y increases **downwards**.
- The **game loop** repeats three jobs every frame: handle events, update the game, draw the screen.
- **KEYDOWN** fires once per key press, so the square moves one step per press.

Activities

1. Run the starter program and label the event loop, the drawing code and the display update.
2. Change the shape's colour, size and starting position.
3. Change how far the shape moves with each press.
4. Add another key (for example WASD) to control the shape.

Code

Complete program

```
import pygame

pygame.init()

WIDTH = 800
HEIGHT = 600
SIZE = 40

screen = pygame.display.set_mode((WIDTH, HEIGHT))
pygame.display.set_caption("Move the square")
clock = pygame.time.Clock()

x = 380
y = 280
running = True

while running:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False
        elif event.type == pygame.KEYDOWN:
            if event.key == pygame.K_LEFT:
                x -= 20
            elif event.key == pygame.K_RIGHT:
                x += 20
            elif event.key == pygame.K_UP:
                y -= 20
            elif event.key == pygame.K_DOWN:
                y += 20
```

```
        screen.fill((30, 30, 40))
        pygame.draw.rect(screen, (80, 200, 120), (x, y, SIZE, SIZE))
        pygame.display.flip()
        clock.tick(60)

    pygame.quit()
```

Checkpoint

The shape moves in all four directions. Learners can name the variables that control horizontal and vertical position.

Support and stretch

- **Support:** Give a partly completed version with some key checks missing.
- **Stretch:** Add WASD controls, or draw a second shape that moves in the opposite direction.

Reflect

What would happen if `screen.fill()` was removed? Predict, then test it.

Save your work
Save this version as `lesson1.py` before you start the next lesson.